Experiment 2: To Perform Data Preprocessing Using Python

Aim: To perform data preprocessing using Python by handling missing values, removing duplicate records, encoding categorical data, and scaling numerical features.

Objectives:

- Load a dataset using Python.
- Identify and handle missing values.
- Remove duplicate records.
- Convert categorical data into numerical form.
- Normalize/standardize numerical data.
- Prepare clean data for further analysis or machine learning.

Software/Tools Required:

- Python 3.x
- Jupyter Notebook / Google Colab
- Pandas
- NumPy
- Scikit-learn

Theory:

Data preprocessing is an important step in data analysis that converts raw **JioHotstar** data into a clean, consistent, and suitable format for further analysis and machine learning. The **JioHotstar** dataset may contain missing values, duplicate records, categorical information such as content type, language, genre, subscription plan, and numerical information such as watch time, age, rating, and subscription amount.

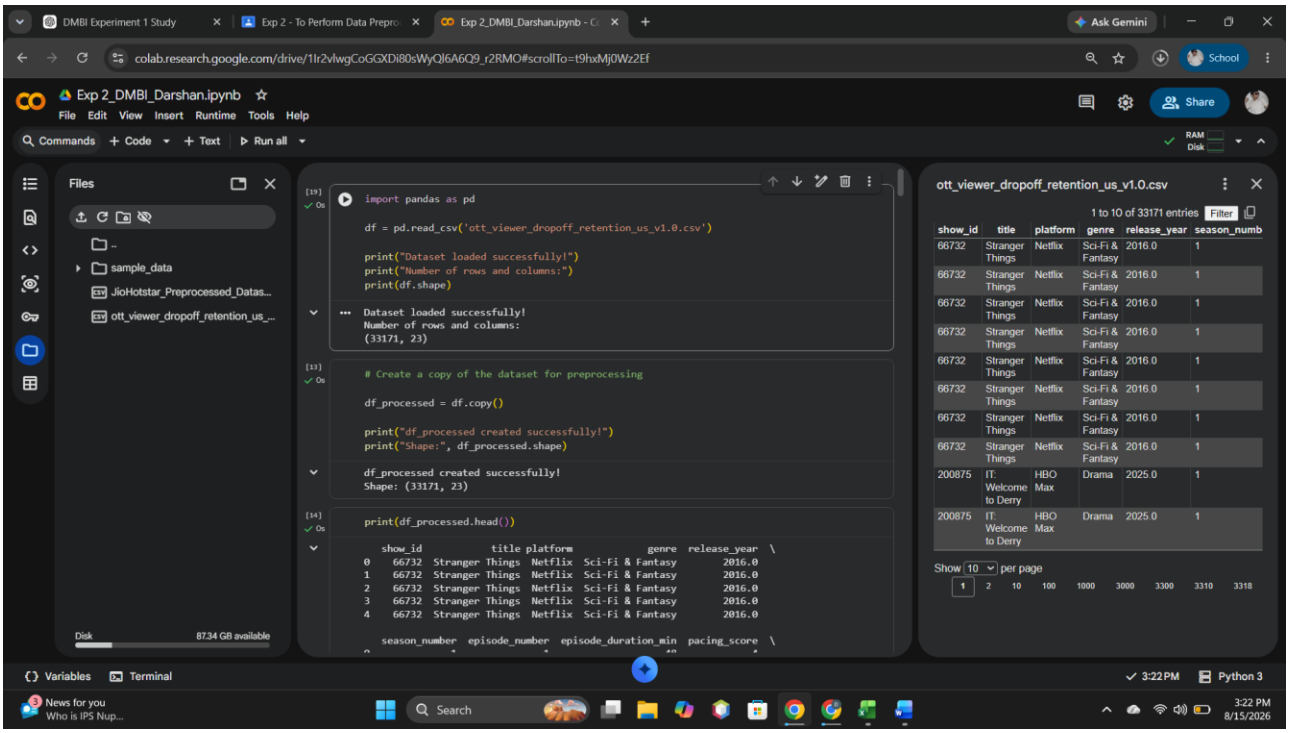
The following preprocessing techniques are applied to the JioHotstar dataset:

1. **Data Cleaning:** Missing or incorrect values in attributes such as age, watch time, rating, or subscription details are identified and handled using suitable methods such as mean, median, or mode.
2. **Removing Duplicates:** Duplicate user or viewing records are removed to avoid repetition and ensure accurate analysis.
3. **Encoding Categorical Data:** Categorical attributes such as gender, language, genre, content type, and subscription plan are converted into numerical values using encoding techniques such as Label Encoding or One-Hot Encoding.

4. **Feature Scaling:** Numerical attributes such as age, watch time, watch percentage, and subscription amount may have different ranges. Standardization or normalization is used to bring these features to a similar scale.
5. **Data Transformation:** New useful features can be created or existing features can be transformed into a suitable format. For example, watch percentage can be converted into watch engagement levels such as Low, Medium, and High.

After preprocessing, the JioHotstar dataset becomes clean, consistent, and ready for further data analysis, visualization, and machine learning applications.

OUTPUTS:




```
# Create a copy of the dataset for preprocessing

df_processed = df.copy()

print("df_processed created successfully!")
print("Shape:", df_processed.shape)
```

```
df_processed created successfully!
Shape: (33171, 23)
```

```
print(df_processed.head())
```

```
   show_id           title platform             genre  release_year  \
0    66732  Stranger Things  Netflix  Sci-Fi & Fantasy        2016.0
1    66732  Stranger Things  Netflix  Sci-Fi & Fantasy        2016.0
2    66732  Stranger Things  Netflix  Sci-Fi & Fantasy        2016.0
3    66732  Stranger Things  Netflix  Sci-Fi & Fantasy        2016.0
4    66732  Stranger Things  Netflix  Sci-Fi & Fantasy        2016.0

   season_number  episode_number  episode_duration_min  pacing_score  \
0              1               1                    48             4
1              1               2                    55             5
2              1               3                    51             4
3              1               4                    50             4
4              1               5                    52             4

   hook_strength  ... pause_count  rewind_count  skip_intro  cognitive_load  \
0              5  ...           3             0           0               9
1              4  ...           3             3           1               5
2              8  ...           4             2           0               9
3              7  ...           4             1           0               7
4              3  ...           3             0           1               7

   attention_required  night_watch_safe  drop_off drop_off_probability  \
0                high                 0         1                0.649
1              medium                 0         0                0.473
2                high                 0         0                0.583
3                high                 0         0                0.520
4                high                 0         1                0.638

   retention_risk  dataset_version
0            high            v1.0
1          medium            v1.0
2          medium            v1.0
3          medium            v1.0
4            high            v1.0

[5 rows x 23 columns]
```

```
# Data Transformation: Create watch engagement levels

df_processed['watch_engagement_level'] = pd.cut(
    df_processed['avg_watch_percentage'],
    bins=[-float('inf'), 40, 70, float('inf')],
    labels=['Low', 'Medium', 'High']
)

print("Watch engagement levels created successfully!")
print(df_processed['watch_engagement_level'].value_counts())
```

```
Watch engagement levels created successfully!
watch_engagement_level
Medium    24844
High       4781
Low        3546
Name: count, dtype: int64
```

```
print("Final Dataset Shape:", df_processed.shape)

print("\nMissing Values After Preprocessing:")
print(df_processed.isnull().sum().sum())

print("\nDuplicate Rows After Preprocessing:")
print(df_processed.duplicated().sum())
```

```
Final Dataset Shape: (33171, 24)

Missing Values After Preprocessing:
0

Duplicate Rows After Preprocessing:
0
```

```
print("First 5 rows of the dataset:")
print(df.head())
```

```
First 5 rows of the dataset:
   show_id           title platform             genre  release_year  \
0    66732  Stranger Things  Netflix  Sci-Fi & Fantasy        2016.0
1    66732  Stranger Things  Netflix  Sci-Fi & Fantasy        2016.0
2    66732  Stranger Things  Netflix  Sci-Fi & Fantasy        2016.0
3    66732  Stranger Things  Netflix  Sci-Fi & Fantasy        2016.0
4    66732  Stranger Things  Netflix  Sci-Fi & Fantasy        2016.0

   season_number  episode_number  episode_duration_min  pacing_score  \
0              1               1                    48             4
1              1               2                    55             5
2              1               3                    51             4
3              1               4                    50             4
4              1               5                    52             4

   hook_strength  ... pause_count  rewind_count  skip_intro  cognitive_load  \
0              5  ...           3             0           0               9
1              4  ...           3             3           1               5
2              8  ...           4             2           0               9
3              7  ...           4             1           0               7
4              3  ...           3             0           1               7

   attention_required  night_watch_safe  drop_off drop_off_probability  \
0                high                 0         1                0.649
1              medium                 0         0                0.473
2                high                 0         0                0.583
3                high                 0         0                0.520
4                high                 0         1                0.638

   retention_risk  dataset_version
0            high            v1.0
1          medium            v1.0
2          medium            v1.0
3          medium            v1.0
4            high            v1.0

[5 rows x 23 columns]
```

```
print("Dataset Information:")
print(df.info())
```

```
Dataset Information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 33171 entries, 0 to 33170
Data columns (total 23 columns):
 #   Column                Non-Null Count  Dtype
---  ------                --------------  -----
 0   show_id               33171 non-null  int64
 1   title                 33171 non-null  object
 2   platform              33171 non-null  object
 3   genre                 33171 non-null  object
 4   release_year          33171 non-null  float64
 5   season_number         33171 non-null  int64
 6   episode_number        33171 non-null  int64
 7   episode_duration_min  33171 non-null  int64
 8   pacing_score          33171 non-null  int64
 9   hook_strength         33171 non-null  int64
 10  dialogue_density      33171 non-null  object
 11  visual_intensity      33171 non-null  int64
 12  avg_watch_percentage  33171 non-null  int64
 13  pause_count           33171 non-null  int64
 14  rewind_count          33171 non-null  int64
 15  skip_intro            33171 non-null  int64
 16  cognitive_load        33171 non-null  int64
 17  attention_required    33171 non-null  object
 18  night_watch_safe      33171 non-null  int64
 19  drop_off              33171 non-null  int64
 20  drop_off_probability  33171 non-null  float64
 21  retention_risk        33171 non-null  object
 22  dataset_version       33171 non-null  object
dtypes: float64(2), int64(14), object(7)
memory usage: 5.8+ MB
None
```

```
print("Missing Values:")
print(df.isnull().sum())
```

```
Missing Values:
show_id                 0
title                   0
platform                0
genre                   0
release_year            0
season_number           0
episode_number          0
episode_duration_min    0
pacing_score            0
hook_strength           0
dialogue_density        0
visual_intensity        0
avg_watch_percentage    0
pause_count             0
rewind_count            0
skip_intro              0
cognitive_load          0
attention_required      0
night_watch_safe        0
drop_off                0
drop_off_probability    0
retention_risk          0
dataset_version         0
dtype: int64
```

```
print("Duplicate Rows:")
print(df.duplicated().sum())
```

```
Duplicate Rows:
0
```

```
print("Categorical Columns:")
print(df.select_dtypes(include='object').columns.tolist())
```

```
Categorical Columns:
['title', 'platform', 'genre', 'dialogue_density', 'attention_required', 'retention_risk', 'dataset_version']
```

```
print("Numerical Columns:")
print(df.select_dtypes(include=['int64', 'float64']).columns.tolist())
```

```
Numerical Columns:
['show_id', 'release_year', 'season_number', 'episode_number', 'episode_duration_min', 'pacing_score', 'hook_
```

```
print("Unique values in categorical columns:\n")

for col in df.select_dtypes(include='object').columns:
    print(f"\n{col}:")
    print(df[col].unique()[:20])
```

```
Unique values in categorical columns:


title:
['Stranger Things' 'IT: Welcome to Derry' 'Spartacus: House of Ashur'
 'Supernatural' 'Man vs Baby' 'Law & Order: Special Victims Unit'
 "Grey's Anatomy" 'Game of Thrones'
 'The Boys from Brazil: Rise of the Bolsonaros' 'The Simpsons'
 'The Rookie' 'NCIS' 'Record of Ragnarok' 'Dexter' 'Pluribus' 'Shameless'
 'Law & Order' 'Family Guy' 'Breaking Bad' 'House']

platform:
['Netflix' 'HBO Max' 'Starz Apple TV Channel' 'Hulu' 'Disney+' 'fuboTV'
 'Apple TV' 'Prime Video' 'Paramount Plus Apple TV Channel ' 'HiDive'
 'Spectrum On Demand' 'Peacock Premium' 'Crunchyroll'
 'MGM+ Amazon Channel' 'Rakuten Viki' 'Britbox Apple TV Channel '
 'OnDemandKorea' 'AMC Plus Apple TV Channel ' 'YouTube TV'
 'Kocowa Amazon Channel']

genre:
['Sci-Fi & Fantasy' 'Drama' 'Comedy' 'Crime' 'Documentary' 'Family'
 'Action & Adventure' 'Animation' 'News' 'Western' 'Mystery' 'Talk'
 'War & Politics' 'Reality' 'Kids' 'Soap']

dialogue_density:
['high' 'low' 'medium']

attention_required:
['high' 'medium']

retention_risk:
['high' 'medium' 'low']

dataset_version:
['v1.0']
```

```
print("Dialogue Density values:")
print(df['dialogue_density'].value_counts())
```

```
Dialogue Density values:
dialogue_density
high      11115
low       11115
medium    10941
Name: count, dtype: int64
```

```
print("Numerical Summary:")
print(df.describe())
```

```
Numerical Summary:
              show_id  release_year  season_number  episode_number  \
count   33171.000000  33171.000000        33171.0    33171.000000
mean    63698.539688   2000.739743            1.0      423.718881
std     65206.256887     15.750470            0.0      557.688460
min        36.000000   1952.000000            1.0        1.000000
25%     15226.000000   1992.000000            1.0       27.000000
50%     35024.000000   2004.000000            1.0      178.000000
75%     80885.000000   2012.000000            1.0      611.000000
max    305062.000000   2025.000000            1.0     2650.000000

       episode_duration_min  pacing_score  hook_strength  visual_intensity  \
count          33171.000000  33171.000000   33171.000000      33171.000000
mean              37.660667      5.397908       5.525158          6.009436
std               22.899741      1.230418       1.730098          1.994370
min                1.000000      3.000000       3.000000          3.000000
25%               24.000000      4.000000       4.000000          4.000000
50%               33.000000      5.000000       6.000000          6.000000
75%               49.000000      6.000000       7.000000          8.000000
max             1225.000000      8.000000      10.000000          9.000000

       avg_watch_percentage   pause_count  rewind_count    skip_intro  \
count          33171.000000  33171.000000  33171.000000  33171.000000
mean              56.805764      3.102921      2.003497      0.493262
std               12.577000      1.866938      1.548391      0.499962
min               13.000000      0.000000      0.000000      0.000000
25%               48.000000      2.000000      1.000000      0.000000
50%               57.000000      3.000000      2.000000      0.000000
75%               66.000000      4.000000      3.000000      1.000000
max              100.000000     10.000000      8.000000      1.000000

       cognitive_load  night_watch_safe      drop_off  drop_off_probability
count    33171.000000      33171.000000  33171.000000          33171.000000
mean         6.196015          0.015887      0.144825              0.478433
std          1.322146          0.125042      0.351930              0.107382
min          4.000000          0.000000      0.000000              0.120000
25%          5.000000          0.000000      0.000000              0.400000
50%          7.000000          0.000000      0.000000              0.471000
75%          7.000000          0.000000      0.000000              0.553000
max          9.000000          1.000000      1.000000              0.831000
```

```
df_processed = df.copy()

print("Copy created successfully!")
print(df_processed.shape)
```

```
Copy created successfully!
(33171, 23)
```

```
df_processed = df.copy()

print("Copy created successfully!")
print(df_processed.shape)
```

```
Copy created successfully!
(33171, 23)
```

+ Code + Text

```
from sklearn.preprocessing import LabelEncoder

label_encoder = LabelEncoder()

categorical_columns = [
    'platform',
    'genre',
    'dialogue_density',
    'attention_required',
    'retention_risk'
]

for col in categorical_columns:
    df_processed[col] = label_encoder.fit_transform(df_processed[col])

print("Categorical data encoded successfully!")
print(df_processed[categorical_columns].head())
```

```
Categorical data encoded successfully!
   platform  genre  dialogue_density  attention_required  retention_risk
0        13     11                 0                   0               0
1        13     11                 1                   1               2
2        13     11                 0                   0               2
3        13     11                 2                   0               2
4        13     11                 1                   0               0
```

```
print("Final Preprocessed Dataset:")
print(df_processed.head())
```

```
Final Preprocessed Dataset:
   show_id           title  platform  genre  release_year  season_number  \
0    66732  Stranger Things        13     11        2016.0              1
1    66732  Stranger Things        13     11        2016.0              1
2    66732  Stranger Things        13     11        2016.0              1
3    66732  Stranger Things        13     11        2016.0              1
4    66732  Stranger Things        13     11        2016.0              1

   episode_number  episode_duration_min  pacing_score  hook_strength  ...  \
0               1              0.451511     -1.136141      -0.303547  ...
1               2              0.757196     -0.323397      -0.881557  ...
2               3              0.582519     -1.136141       1.430485  ...
3               4              0.538850     -1.136141       0.852475  ...
4               5              0.626188     -1.136141      -1.459568  ...

   pause_count  rewind_count  skip_intro  cognitive_load  attention_required  \
0    -0.055129     -1.293941           0        2.120815                   0
1    -0.055129      0.643583           1       -0.904614                   1
2     0.480515     -0.002259           0        2.120815                   0
3     0.480515     -0.648100           0        0.608100                   0
4    -0.055129     -1.293941           1        0.608100                   0

   night_watch_safe  drop_off  drop_off_probability  retention_risk  \
0                 0         1              1.588437               0
1                 0         0             -0.050595               2
2                 0         0              0.973800               2
3                 0         0              0.387101               2
4                 0         1              1.485997               0

   dataset_version
0             v1.0
1             v1.0
2             v1.0
3             v1.0
4             v1.0

[5 rows x 23 columns]
```

```
print("Final Dataset Shape:", df_processed.shape)

print("\nMissing Values After Preprocessing:")
print(df_processed.isnull().sum().sum())

print("\nDuplicate Rows After Preprocessing:")
print(df_processed.duplicated().sum())
```

```
Final Dataset Shape: (33171, 23)

Missing Values After Preprocessing:
0

Duplicate Rows After Preprocessing:
0
```

```
df_processed.to_csv('JioHotstar_Preprocessed_Dataset.csv', index=False)

print("Processed dataset saved successfully!")
```

```
Processed dataset saved successfully!
```

```
[ ]
from google.colab import files

files.download('JioHotstar_Preprocessed_Dataset.csv')
```

```
[17]
✓ 1s
# Save the preprocessed dataset

df_processed.to_csv(
    'JioHotstar_Preprocessed_Dataset.csv',
    index=False
)

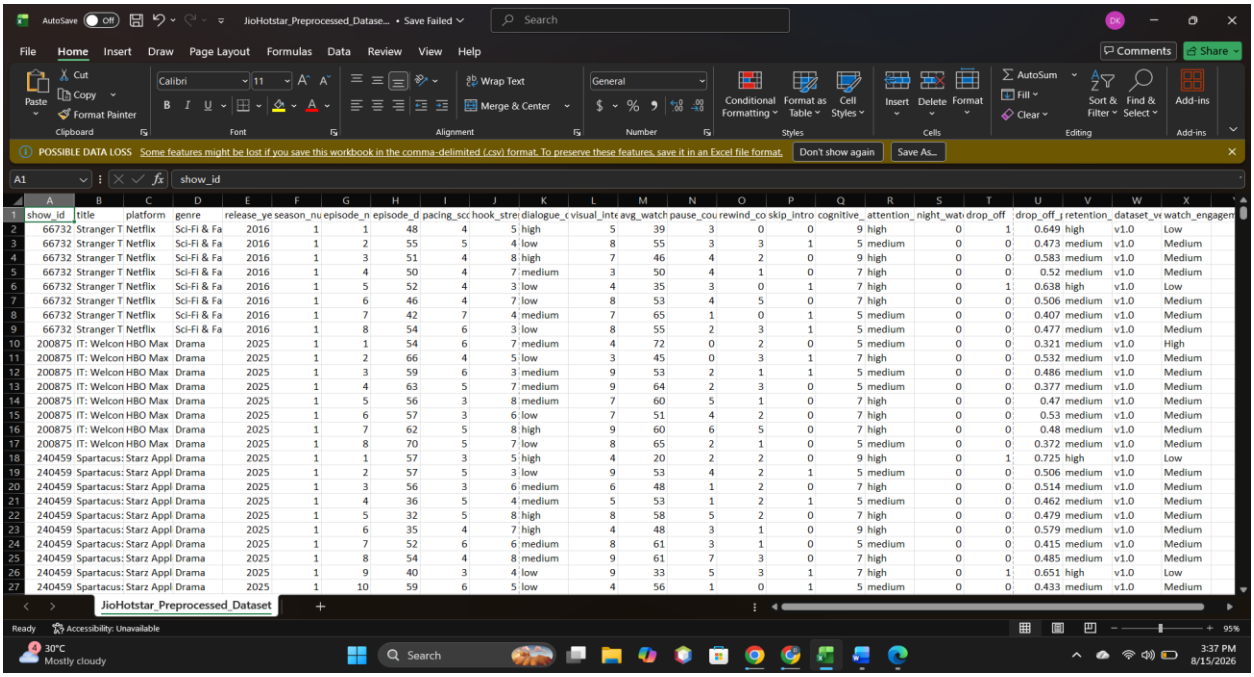
print("Processed dataset saved successfully!")
```

```
Processed dataset saved successfully!
```

```
[18]
✓ 0s
from google.colab import files

files.download('JioHotstar_Preprocessed_Dataset.csv')
```

I have added screenshot, for my final refined Data Set for my problem statement (i.e Jiohotar) below:



show_id	title	platform	genre	release_ye	season_nu	episode_n	episode_d	pacing_sc	hook_stre	dialogue_c	visual_inte	avg_watch	pause_cou	rewind_co	skip_intro	cognitive_	attention_	night_wat	drop_off	drop_off_r	retention_	dataset_ve	watch_engagem
66732	Stranger T	Netflix	Sci-Fi & Fa	2016	1	1	48	4	5	high	5	39	3	0	0	9	high	0	1	0.649	high	v1.0	Low
66732	Stranger T	Netflix	Sci-Fi & Fa	2016	1	2	55	5	4	low	8	55	3	3	1	5	medium	0	0	0.473	medium	v1.0	Medium
66732	Stranger T	Netflix	Sci-Fi & Fa	2016	1	3	51	4	8	high	7	46	4	2	0	9	high	0	0	0.583	medium	v1.0	Medium
66732	Stranger T	Netflix	Sci-Fi & Fa	2016	1	4	50	4	7	medium	3	50	4	1	0	7	high	0	0	0.52	medium	v1.0	Medium
66732	Stranger T	Netflix	Sci-Fi & Fa	2016	1	5	52	4	3	low	4	35	3	0	1	7	high	0	1	0.638	high	v1.0	Low
66732	Stranger T	Netflix	Sci-Fi & Fa	2016	1	6	46	4	7	low	8	53	4	5	0	7	high	0	0	0.506	medium	v1.0	Medium
66732	Stranger T	Netflix	Sci-Fi & Fa	2016	1	7	42	7	4	medium	7	65	1	0	1	5	medium	0	0	0.407	medium	v1.0	Medium
66732	Stranger T	Netflix	Sci-Fi & Fa	2016	1	8	54	6	3	low	8	55	2	3	1	5	medium	0	0	0.477	medium	v1.0	Medium
200875	IT: Welcon	HBO Max	Drama	2025	1	1	54	6	7	medium	4	72	0	2	0	5	medium	0	0	0.321	medium	v1.0	High
200875	IT: Welcon	HBO Max	Drama	2025	1	2	66	4	5	low	3	45	0	3	1	7	high	0	0	0.532	medium	v1.0	Medium
200875	IT: Welcon	HBO Max	Drama	2025	1	3	59	6	3	medium	9	53	2	1	1	5	medium	0	0	0.486	medium	v1.0	Medium
200875	IT: Welcon	HBO Max	Drama	2025	1	4	63	5	7	medium	9	64	2	3	0	5	medium	0	0	0.377	medium	v1.0	Medium
200875	IT: Welcon	HBO Max	Drama	2025	1	5	56	3	8	medium	7	60	5	1	0	7	high	0	0	0.47	medium	v1.0	Medium
200875	IT: Welcon	HBO Max	Drama	2025	1	6	57	3	6	low	7	51	4	2	0	7	high	0	0	0.53	medium	v1.0	Medium
200875	IT: Welcon	HBO Max	Drama	2025	1	7	62	5	8	high	9	60	6	5	0	7	high	0	0	0.48	medium	v1.0	Medium
200875	IT: Welcon	HBO Max	Drama	2025	1	8	70	5	7	low	8	65	2	1	0	5	medium	0	0	0.372	medium	v1.0	Medium
240459	Spartacus:	Starz Appl	Drama	2025	1	1	57	3	5	high	4	20	2	2	0	9	high	0	1	0.725	high	v1.0	Low
240459	Spartacus:	Starz Appl	Drama	2025	1	2	57	5	3	low	9	53	4	2	1	5	medium	0	0	0.506	medium	v1.0	Medium
240459	Spartacus:	Starz Appl	Drama	2025	1	3	56	3	6	medium	6	48	1	2	0	7	high	0	0	0.514	medium	v1.0	Medium
240459	Spartacus:	Starz Appl	Drama	2025	1	4	36	5	4	medium	5	53	1	2	1	5	medium	0	0	0.462	medium	v1.0	Medium
240459	Spartacus:	Starz Appl	Drama	2025	1	5	32	5	8	high	8	58	5	2	0	7	high	0	0	0.479	medium	v1.0	Medium
240459	Spartacus:	Starz Appl	Drama	2025	1	6	35	4	7	high	4	48	3	1	0	9	high	0	0	0.579	medium	v1.0	Medium
240459	Spartacus:	Starz Appl	Drama	2025	1	7	52	6	6	medium	8	61	3	1	0	5	medium	0	0	0.415	medium	v1.0	Medium
240459	Spartacus:	Starz Appl	Drama	2025	1	8	54	4	8	medium	9	61	7	3	0	7	high	0	0	0.485	medium	v1.0	Medium
240459	Spartacus:	Starz Appl	Drama	2025	1	9	40	3	4	low	9	33	5	3	1	7	high	0	1	0.651	high	v1.0	Low
240459	Spartacus:	Starz Appl	Drama	2025	1	10	59	6	5	low	4	56	1	0	1	5	medium	0	0	0.433	medium	v1.0	Medium

Result:

Thus, the JioHotstar dataset was successfully preprocessed using Python by handling missing values, removing duplicate records, encoding categorical data, scaling numerical features, and transforming the data into a suitable format. The resulting clean dataset is ready for further data analysis, visualization, and machine learning.